Modern Data Science Project Guide

This document explains each proposed project in the data science roadmap: what it teaches, what to build, what to focus on, what to avoid, and how to judge whether the project is truly portfolio-ready. The sequence is intentionally progressive, moving from analysis and statistics into machine learning and then into more modern applied data science projects.[cite:53] [cite:57]

The projects are especially chosen to support both freelancing and future company roles, because they show not only coding ability but also business thinking, communication, and practical analytical judgment.[cite:54] [cite:57]

Project 1: EDA and KPI analysis

Goal

Build a full exploratory analysis on a business dataset such as e-commerce, sales, marketplace activity, or customer transactions. The point is to prove that raw data can be turned into useful questions, insights, and recommendations.[cite:56] [cite:57]

What you learn

- Data cleaning.
- Exploratory analysis.
- Groupby logic in pandas.
- Chart choice.
- Basic KPI design.
- Insight communication.

What to build

Use a dataset with dates, categories, users/customers, and revenue or activity signals. Create a notebook that covers data quality, key KPIs, segmentation, trends over time, and recommendations.

What to focus on

- Write the business questions first.
- Show data issues early.
- Use only charts that answer a question.
- End with 5 to 10 real findings.

What to avoid

- Too many plots without explanation.
- No business summary.
- Ignoring missing values.
- Using a dataset so small that no serious pattern exists.

How to know you did it well

- A non-technical person could read the notebook and understand the situation.
- Your charts are clean and purposeful.
- Your conclusions are specific, not generic.

Project 2: Retention, cohort, or funnel analysis

Goal

Analyze user behavior over time to understand acquisition quality, retention, dropout points, or conversion funnel weakness. This type of work is very valuable in product and growth-focused teams.[cite:57]

What you learn

- Cohort analysis.
- Funnel metrics.
- User segmentation.
- Product thinking.
- Metric interpretation.

What to build

Choose a dataset with user events, signups, sessions, orders, or subscriptions. Build retention tables, funnel drop-off analysis, and segment comparisons.

What to focus on

- Define the user journey clearly.
- Explain why users drop or retain.
- Compare segments like source, device, location, or user type.
- Recommend concrete next actions.

What to avoid

- Treating all users as one group.
- Mixing definitions of retention.
- Plotting percentages without sample-size context.

How to know you did it well

- You can explain the funnel and retention logic clearly.
- You identify the biggest behavioral bottleneck.
- You propose realistic product or business interventions.

Project 3: A/B test or experimentation analysis

Goal

Study an experiment and determine whether a change improved a target metric in a trustworthy way. This project proves statistical maturity, not just plotting skill.[cite:50]

What you learn

- Hypothesis testing.
- Confidence intervals.
- Practical significance.
- Guardrail metrics.
- Segment analysis.

What to build

Use an A/B testing dataset or simulate one carefully. Analyze conversion rates or engagement metrics, check whether the difference is meaningful, and discuss risk and rollout implications.

What to focus on

- State the null and alternative hypotheses.
- Define the primary metric and guardrails.
- Distinguish significance from business value.
- Check whether effects differ across segments.

What to avoid

- Reporting only p-values.
- Ignoring sample size and power.
- Pretending one experiment proves everything.

How to know you did it well

- You can answer: “Should the company ship this change, and why?”
- You discuss uncertainty honestly.
- Your analysis feels decision-oriented.

Project 4: Customer churn prediction

Goal

Predict which customers are likely to stop using a service or stop purchasing. This is one of the best entry-level ML projects because it connects clearly to business value.[cite:53] [cite:57]

What you learn

- Binary classification.
- Feature engineering.
- Train/test split and evaluation.
- Class imbalance handling.
- Error analysis.
- Business interpretation of predictions.

What to build

Use a churn dataset or a subscription/customer activity dataset. Build a baseline model, improve it, and analyze which features matter.

What to focus on

- Start with a simple baseline.
- Use business-relevant metrics, not accuracy alone.
- Investigate false positives and false negatives.
- Translate predictions into interventions.

What to avoid

- Leakage from future data.
- Using only accuracy on imbalanced classes.
- Claiming the model is useful without a retention strategy.

How to know you did it well

- You can explain what churn means operationally.
- You compare multiple models properly.
- You can suggest how the business should act on the scores.

Project 5: Price prediction or demand forecasting

Goal

Predict a continuous target such as price, sales volume, or demand. This teaches regression thinking and business forecasting discipline.[cite:53]

What you learn

- Regression workflows.
- Feature engineering for continuous outcomes.
- MAE/RMSE evaluation.
- Forecasting basics and seasonality awareness.
- Residual analysis.

What to build

Choose a housing, retail, ride-hailing, inventory, or food-demand dataset. Build a regression model or forecasting baseline and evaluate where it works and fails.

What to focus on

- Understand the target distribution.
- Engineer sensible time or category features.
- Compare naive and stronger baselines.
- Explain uncertainty and operational use.

What to avoid

- Ignoring seasonality or temporal leakage.
- Reporting only one error metric.
- Pretending forecasts are exact.

How to know you did it well

- You can explain when the model is useful and when it breaks.
- Your evaluation reflects real business constraints.
- Your visuals show actual vs predicted cleanly.

Project 6: Recommendation or ranking prototype

Goal

Build a system that suggests items, jobs, products, or content to users. Recommendation is a strong modern specialization because it combines analytics, ML, and product understanding. [cite:51] [cite:57]

What you learn

- Similarity-based systems.
- Collaborative filtering basics.
- Feature-based recommendations.
- Ranking logic.
- Offline evaluation limitations.

What to build

Use movie, e-commerce, jobs, or food preference data. Build a simple recommender and then improve it using user/item features or interaction patterns.

What to focus on

- Define what “good recommendation” means.
- Distinguish retrieval from ranking.
- Handle cold-start issues.
- Explain practical deployment concerns.

What to avoid

- Claiming the recommender is production-ready just because it returns items.
- Ignoring popularity bias.
- Using only one evaluation perspective.

How to know you did it well

- The recommendation logic is clear.
- You discuss trade-offs like relevance, diversity, and cold start.
- You connect the project to product goals.

Project 7: Text analytics or support-ticket intelligence

Goal

Analyze text data to extract patterns, classify cases, or support business operations. Practical NLP is a major part of modern data science because many organizations sit on large amounts of unstructured text.[cite:51] [cite:58]

What you learn

- Text cleaning.
- Vectorization or embeddings.
- Topic discovery.
- Classification.
- Practical NLP evaluation.

What to build

Use support tickets, reviews, survey responses, or comments. Build a project that classifies tickets, extracts themes, or identifies pain points.

What to focus on

- Define a useful business output.
- Compare simple baselines with stronger methods.
- Show how the analysis would help operations or support teams.
- Be careful about ambiguous labels.

What to avoid

- Treating NLP as magic.
- Ignoring annotation quality.
- Building a complex model with no operational value.

How to know you did it well

- Your project produces actionable categories, themes, or triage ideas.
- You explain limitations and possible human-review steps.
- The result feels useful beyond the notebook.

Project 8: Product analytics case study

Goal

Solve a realistic business/product question and write a decision memo. This project proves the part of data science that is hardest to automate: judgment, prioritization, and communication. [cite:56] [cite:57]

What you learn

- Problem framing.
- Metric design.
- Product thinking.
- Decision communication.
- Trade-off explanation.

What to build

Create a case such as declining conversion, poor retention, marketplace imbalance, or drop in order frequency. Use analysis plus a final memo with recommendations and risks.

What to focus on

- Structure your thinking before analysis.
- Define north-star and guardrail metrics.
- Make recommendations prioritized and testable.
- Tie every insight to action.

What to avoid

- Reporting numbers with no implication.
- Acting like the data answers everything.
- Forgetting alternative explanations.

How to know you did it well

- The memo reads like something a manager could use.
- Your recommendations are clear and ordered.
- You show both confidence and uncertainty.

Project 9: End-to-end capstone

Goal

Combine data acquisition, cleaning, analysis, modeling, interpretation, and communication in one polished project. This is your strongest portfolio piece.

What you learn

- End-to-end project execution.
- Scope management.
- Reproducible structure.
- Professional presentation.
- Integration of several data science skills.

What to build

Choose one domain and solve a realistic problem from raw data to final report. Add a clean repo, readme, visual outputs, and possibly a simple dashboard or app.

What to focus on

- Professional structure.
- Clear problem statement.
- Justified methods.
- Strong conclusions and limitations.
- Clean communication.

What to avoid

- Making it too broad.
- Overengineering the modeling part.
- Publishing without cleanup and narrative.

How to know you did it well

- It looks like work that could be shown to a recruiter or client immediately.
- Every file in the project has a purpose.
- You can explain it verbally in 5 minutes and technically in 20.

Best order to build them

The recommended order is:

1. EDA and KPI analysis.
2. Retention or funnel analysis.
3. A/B test analysis.
4. Churn prediction.
5. Price prediction or forecasting.
6. Recommendation system.
7. Text analytics project.
8. Product analytics case study.
9. End-to-end capstone.

This order starts with decision-focused analysis, then adds statistical maturity, then classical ML, then modern specialization projects that show stronger market value.[cite:50] [cite:53] [cite:57]

Portfolio standards for every project

Every serious project should include:

- A clear problem statement.
- Dataset description and data-quality notes.
- Methodology section.
- Results and interpretation.
- Limitations.
- Next steps.
- Clean readme.
- Visual outputs.

- Short business summary.

Career-oriented guidance for data science increasingly stresses portfolio quality, job prep, and communication, so project presentation matters almost as much as model quality when preparing for freelance work or hiring processes.[cite:54]

What makes these projects hard to replace with AI alone

The durable value is not only in writing code. It is in choosing the right problem, checking data validity, selecting metrics, preventing leakage, interpreting uncertainty, communicating trade-offs, and turning outputs into decisions.[cite:56] [cite:57]

That is why the strongest projects are not simply “train a model.” They are business or product cases where analysis, scientific reasoning, and communication matter as much as the algorithm.[cite:51] [cite:57]